

MonEquip Data Warehouse and Analytical Insights (2018–2020)

Data Cleaning • Star Schema Design • Key Business Findings

Data Cleaning

Null Value Problem

Category

Null description
→ updated to "Others"

Duplication Problem

Customer

Customer ID 52 repeated
→ kept one record

Inconsistent Values

Customer

M/F/Male/Female
→ Male/Female

Inconsistent Values

Equipment

HITACHI
→ Hitachi

Inconsistent Values

CustomerType

business → Business
(duplicate removed)

Relationship Problems

Hire

Removed invalid IDs
(301–304)

Incorrect Values

Sales

Fixed wrong quantity
for Sales ID 151

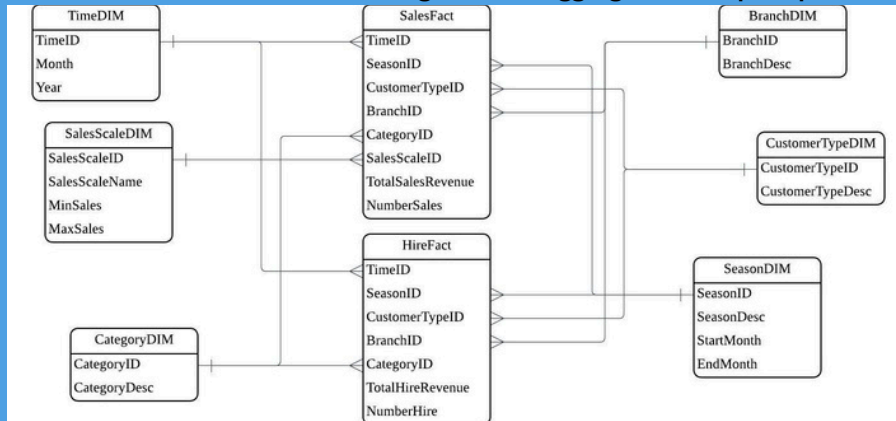
Calculation Errors

Hire

Recomputed 165 records with incorrect hire totals

Star Schema Design

Each record summarises total revenue at a high level of aggregation for quick performance analysis



Store aggregated numeric measures for analysis

Surrogate IDs accurately link facts and dimensions in the aggregated schema

SCD: Type 1 keeps only current values for simplicity

Key Business Findings

Customer Type

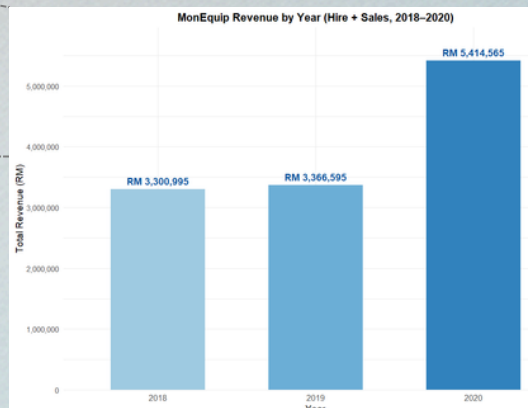
- Business average: RM 28 K
- Individual average: RM 25 K

Category Performance

- Highest: Earthmoving & Landscaping
- Lowest: Site Equipment & Plumbing

Seasonal Trend

- Highest: Spring & Winter
- Lowest: Autumn



Revenue Source

- Sales: 96–98%
- Hire: 2–4%

Sales Scale

- High value transactions: 97%
- Medium value transactions: 3%
- Low value transactions: 0%

Branch Performance

- Highest: Clayton
- Lowest: Cheltenham